جامعة الحسين التقنية
Al Hussein Technical University

Business Process Support

Fall 2025 - 2026

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Activity 1:

1.

City Center is a computer store placed in Jordan that wants to introduce new mechanical gaming keyboard into the market, so the store asked for consultation from NextStep Solutions to evaluate market entry for three keyboards based on historical market data, and they want to see if these keyboards are acceptable by the customers, how future demand will be, to see which is the perfect price should be to have high demand, and which keyboards are most likely must be introduced.

- Razer Huntsman: Amazon - Razer Huntsman
- HyperX Alloy: Amazon - HyperX Alloy
- Redragon K552P: Amazon - Redragon K552P

The target audience for this case are:

- City center sales team: Responsible for understanding the expected demand, pricing behavior, best keyboard.
- City center decision makers: Responsible for deciding if all three keyboards should be introduced or subset of them and deciding when they will be introduced.
- NextStep data analysts: Responsible for data analysis, predictive analysis, visualize analysis, and building the overall solution.

User Requirements:

- Users must be able to understand the overall design, and compare performance, demand, future demand of three keyboards in order to decide which keyboard should be introduced.
- Users must be able to understand brand strength, pricing, and discount effect on customer demand in order to make decisions about pricing.

System Requirements:

- System must have a clean analysis and visualization of sales data including demand, rankings, and performance in order to achieve useful insights.
- System must support forecasting for the next 12 months with an interactive dashboard including future demand.

The data collection process will be using an appropriate tool to collect data from Amazon, as we will use Helium which is an e-commerce data collection tool that offers historical data for the three mechanical gaming keyboards from Razer, Redragon, and HyperX that includes Sales Rank, New Price, List Price, and Time records.

After collecting the data, we will load the data into Power BI, as there will be data preprocessing in Power BI using Power Query Editor, as the data cleaning like handling

missing values, and making the data start from a specific data will be held there. Also, we added new columns and measures that are needed to produce the full analytical results.

After the preprocessing and cleaning step, we will do predictive analysis using forecasting, and it will be used because the dataset has a timestamp and City Center goal is not only to analyze past performance but also to predict future demand by customers, as we will do forecasting using the wizard way in Power BI.

Also, we will use Power BI to make the data visualization step, as we will build dashboards that answers the real City Center goal as these dashboards will include future and historical demand, brand comparisons in terms of demand, and price, the best keyboard performance, averages of price, demand and sales rank, and the change in the new and list price. They will be visualized perfectly for decision makers to help them make an effective decision.

The solution is designed to be easy to use by adding interactive and good design elements such as slicers as everyone can understand and use the overall solution as the dashboards will be clear and straightforward that make audience fully understand these dashboards.

2.

After designing the solution plan, we implemented the data science solution using analytical tools to support the overall goal of City Center.

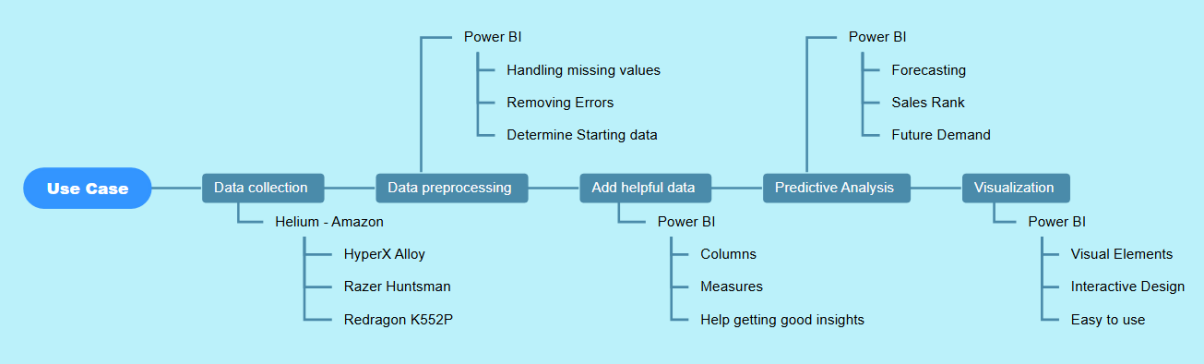


Figure 1: The Implementation Flow

1. Data Collection using Helium:

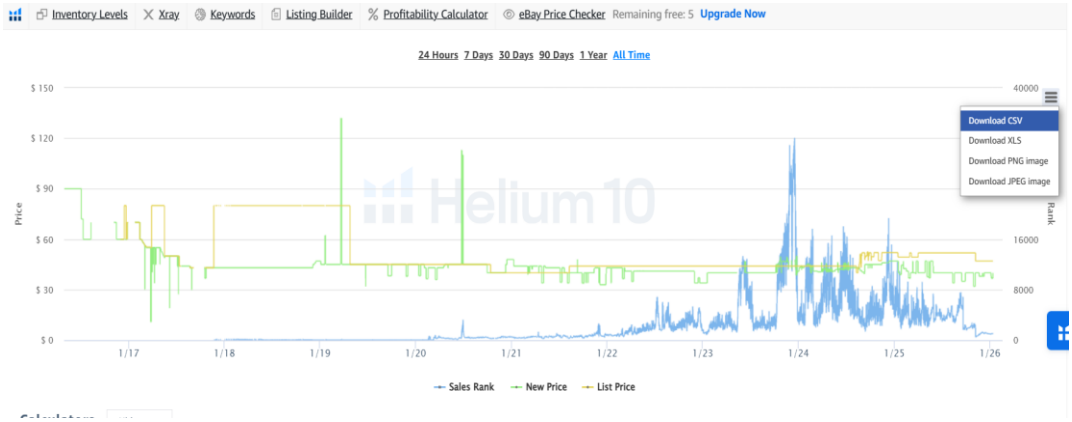


Figure 2: The process of downloading data from Helium.

Figure 3 shows how we collected the data from amazon using Helium, as we did the same process for all three keyboards and the original data starts from different dates and include missing values that was handled in the cleaning step.

2. Data Preprocessing and Cleaning:

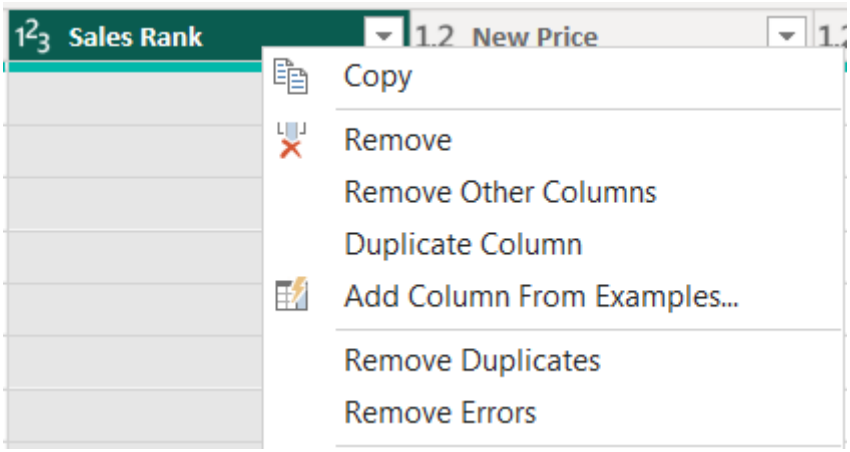


Figure 3: Removing Errors from The Data

Figure 4 shows the process of removing errors from the data, as we did that for some columns by pressing Remove Errors. Remove duplicates weren't needed because of course list price, new price, date, and sales rank can have duplicates.

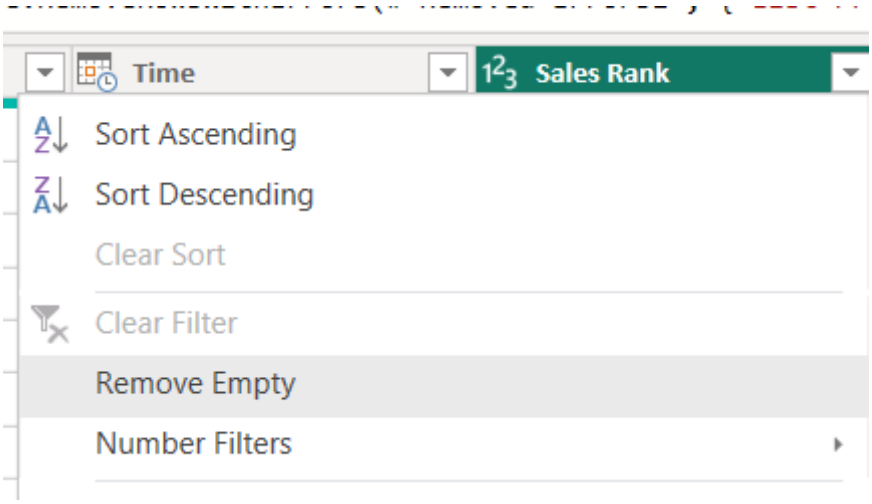


Figure 4: Removing Empty Rows

Figure 5 shows the process or removing empty and missing values in the data, as we preferred that over filling missing values because we already have a very huge data and only few missing values.

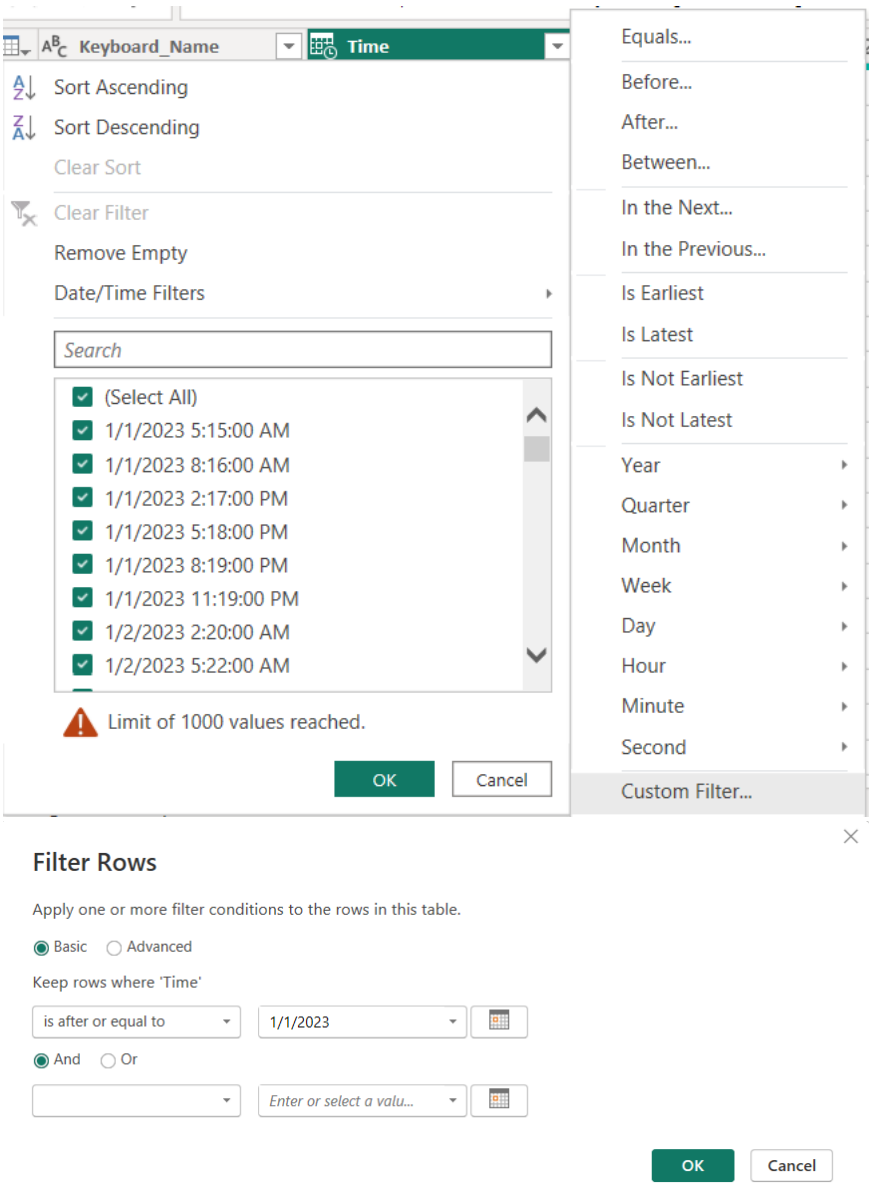


Figure 5: The process of choosing starting date.

Figure 6 shows the process of choosing a date for the data to be started from, as we choose 1/1/2023 as our starting point.

3. Adding Columns and Measures:

```
Brand =
SWITCH(
    TRUE(),
    'Keyboards'[Keyboard_Name] = "HyperX Alloy", "HyperX",
    'Keyboards'[Keyboard_Name] = "Razer Huntsman", "Razer",
    'Keyboards'[Keyboard_Name] = "Redragon K552P", "Redragon"
)
```

Figure 6: Brand Column

```
Brand_ID =
SWITCH(
    TRUE(),
    'Keyboards'[Brand] = "Razer", 1,
    'Keyboards'[Brand] = "HyperX", 2,
    'Keyboards'[Brand] = "Redragon", 3
)
```

Figure 7: Brand_ID Column

```
Demand =
SWITCH(
    TRUE(),
    'Keyboards'[Sales Rank] <= 1000, "High",
    'Keyboards'[Sales Rank] <= 3000, "Medium",
    "Low"
)
```

Figure 8: Demand Column

```
Demand Score =
SWITCH(
    TRUE(),
    'Keyboards'[Demand] = "High", 3,
    'Keyboards'[Demand] = "Medium", 2,
    1
)
```

Figure 9: Demand Score Column

```
New Price (JOD) = Keyboards[New Price] * 0.7095
```

Figure 10: New Price in JOD

```
List Price (JOD) = Keyboards[List Price] * 0.7095
```

Figure 11: List Price in JOD

```
Discount (JOD) = Keyboards[List Price (JOD)] - Keyboards[New Price (JOD)]
```

Figure 12: Discount or Price Change in JOD

Column	Description.
Brand	Specifying keyboard brand.
Brand_ID	Unique id to identify the brand.
Demand	Demand level based on sales rank (High, Medium, Low).
Demand Score	Demand level but numerical to see avg demand for each keyboard (High = 3, Medium = 2, Low = 1).
New Price (JOD)	Selling price in JOD because City Center is placed in Jordan.
List Price (JOD)	Original price in JOD because City Center is placed in Jordan.
Discount (JOD)	Difference between new and list price to see if discount matter to achieve higher demand.

```
1  Avg Demand Score = CALCULATE(AVERAGE(Keyboards[Demand Score]))
```

Figure 13: Avg Demand Score for Each Keyboard

```
1  Keyboard Rank =
2  RANKX(
3      ALL(Keyboards[Keyboard_Name]),
4      [Avg Demand Score]
5  )
```

Figure 14: Keyboard Rank from 1 to 3

Measure	Description
Avg Demand Score	Calculate the average demand score for each keyboard in order to use it in another measure
Keyboard Rank	It returns the keyboard rank based on the average demand score

4. Predictive Analysis using Forecasting:

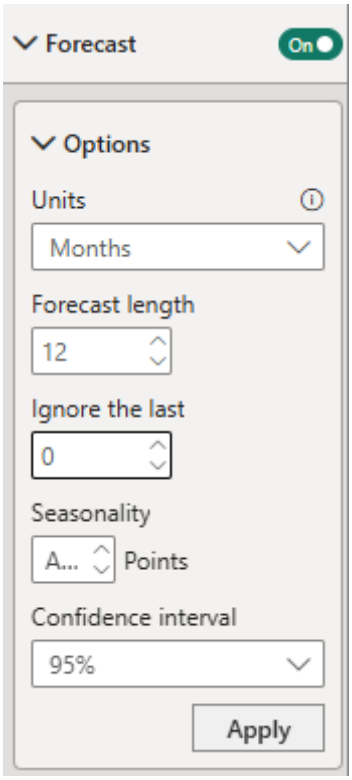


Figure 15: Wizard Forecasting

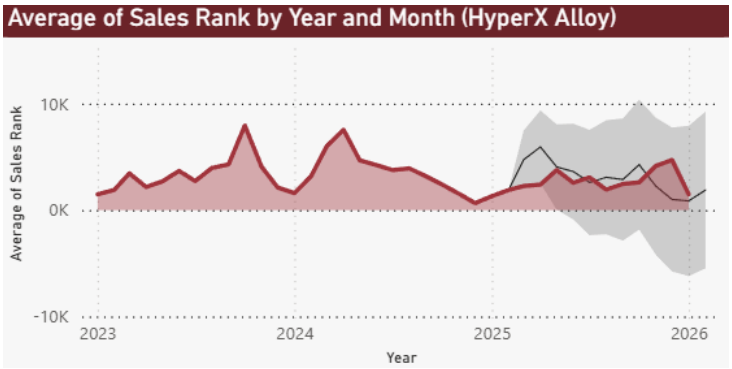


Figure 16: HyperX Alloy Forecasting Performance

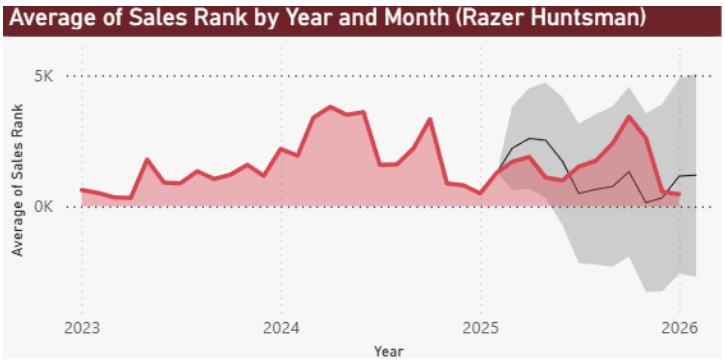


Figure 17: Razer Huntsman Forecasting Performance

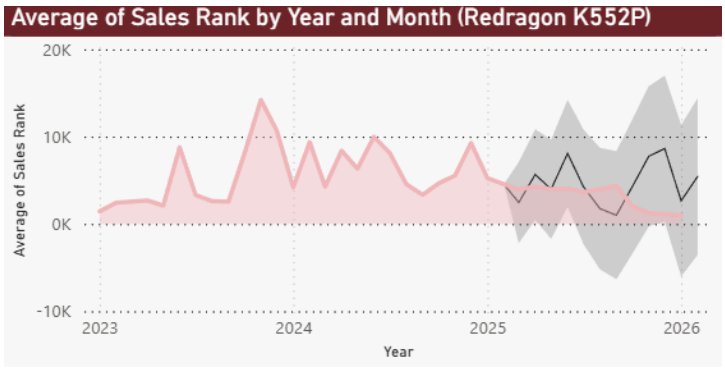


Figure 18: Redragon 552P Forecasting Performance

The figures 17, 18, and 19 show the actual sales rank vs the forecasted or predicted sales rank by ignoring the last 11 months.

5. Visualization:

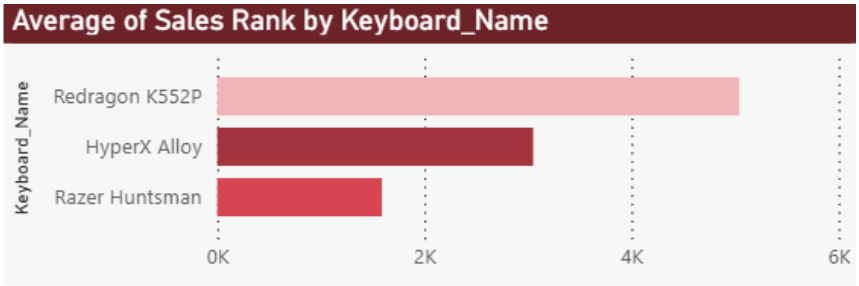


Figure 19: Customers Demand 2

Figure 20 shows City Center two bar charts about how much is every keyboard acceptable by the customers which is indicated by seeing the lowest average sales rank.

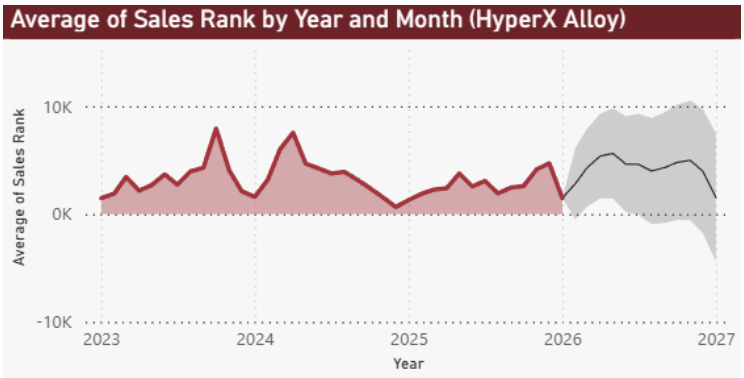


Figure 20: HyperX Alloy Forecsating

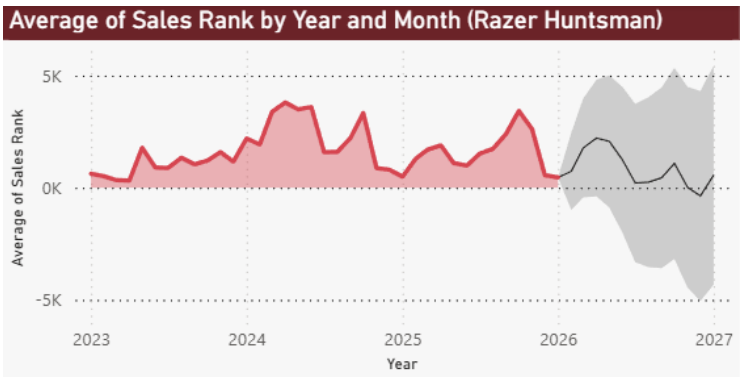


Figure 21: Razer Huntsman Forecasting

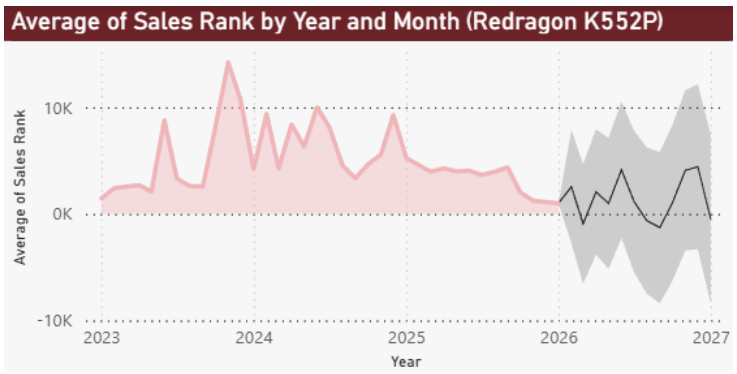


Figure 22: Redragon K552P Forecasting

Figure 21, 22, 23 shows a line chart predicting the future demand and how it will be changed to see if these keyboards are worth being introduced or not.

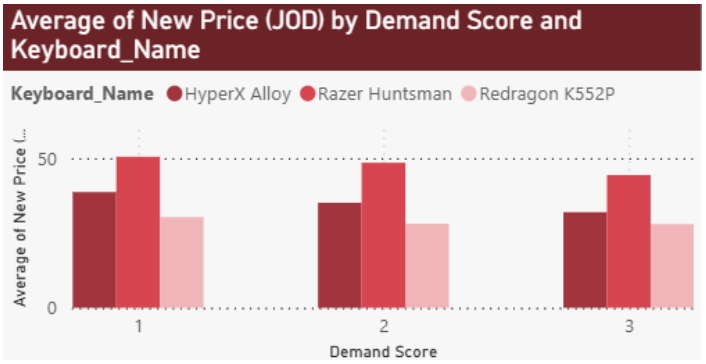


Figure 23: Column Chart to See Average New Price by Demand Score

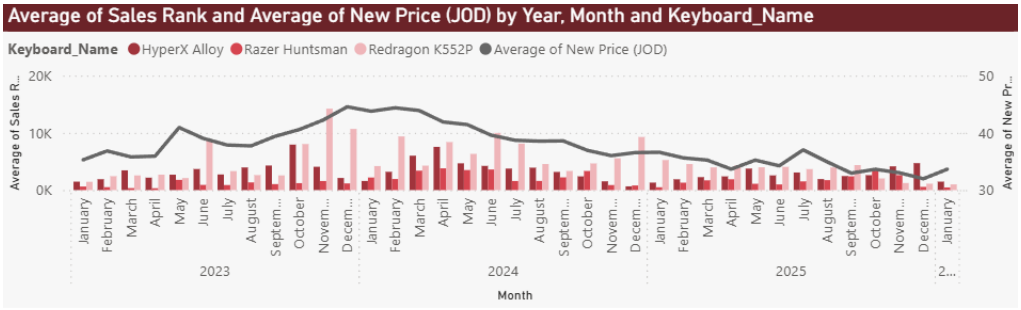


Figure 24: Average New Price and Sales Rank Over Time

Figure 24 shows insights about the perfect price that the keyboard should be introduced to achieve higher demand, see the relationship between price and demand, and see brand influence, but figure 25 shows the same as figure 24 but with an additional thing which is peak months as it is the time that has the highest demand and at which price.

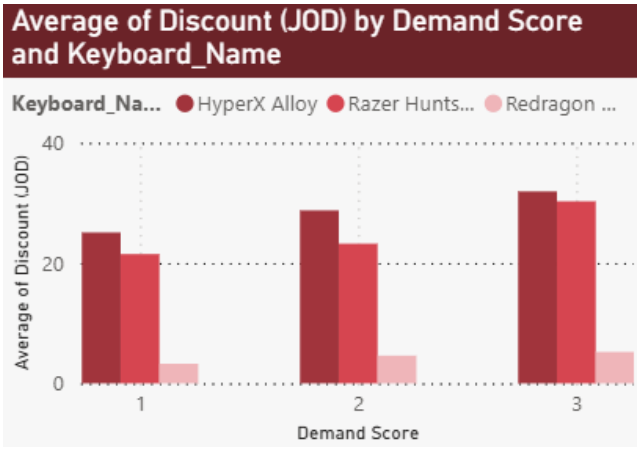


Figure 25: Column Chart to See Discount Effect on Demand

Figure 26 shows the discount effect on demand, as we can see that higher discounts help to achieve higher demand.



Figure 26: Cards and Slicer

Figure 27 shows a summary of the products including average sales rank, new price, demand score, and discount. Also, it shows a slicer for interactivity.

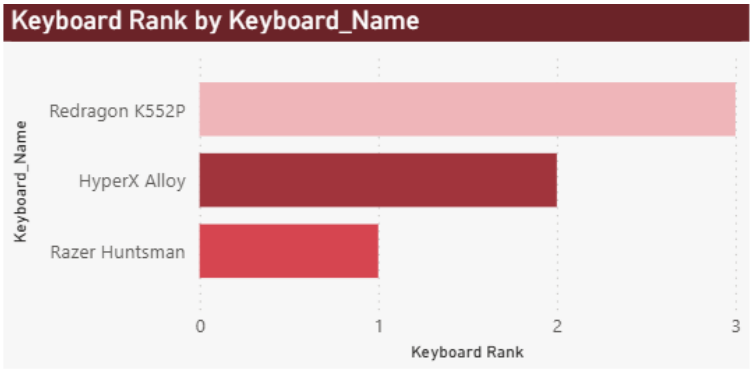


Figure 27: Keyboard Rank

Figure 28 shows the keyboard rank, as it was calculated by converting the sales rank into (high, medium, low), then converting this into demand score (3 = high, 2 =

medium, 1 = low) then by using RankX we calculated the rank for each keyboard based on the average demand score as 3 shows higher rank while 1 shows lower rank.

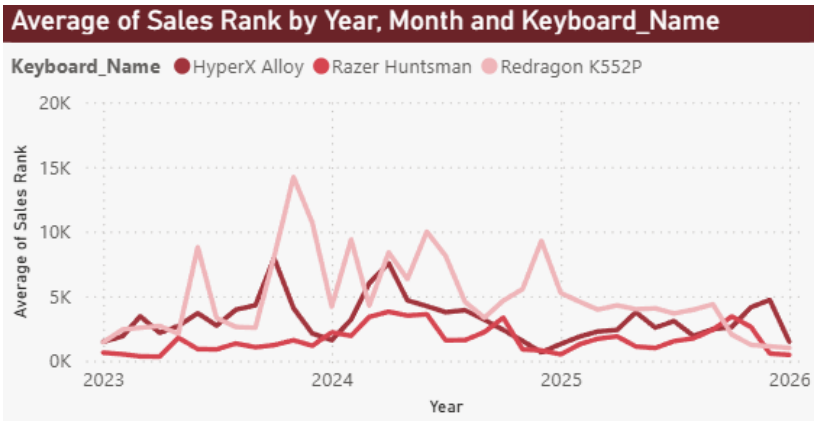


Figure 28: Sales Rank Comparison Over the Time

Figure 29 shows the sales rank for each keyboard over the months and years as we can see from this chart the stability of the sales rank for each keyboard.

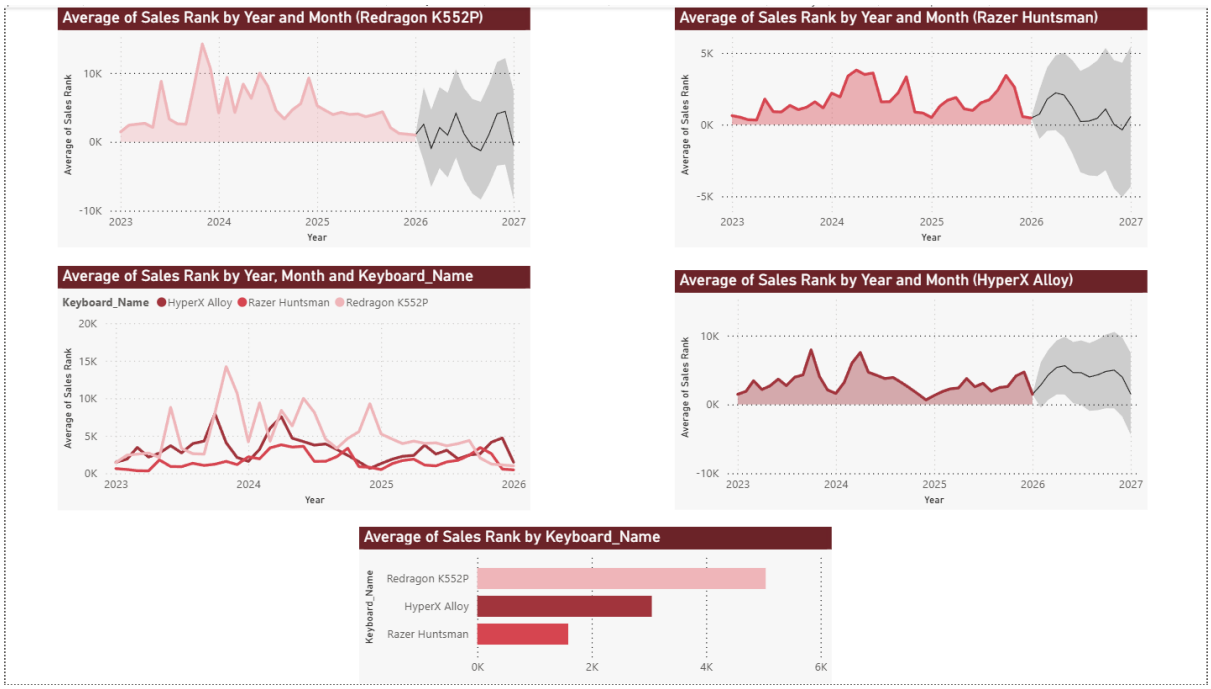


Figure 29:Final Dashboard 1

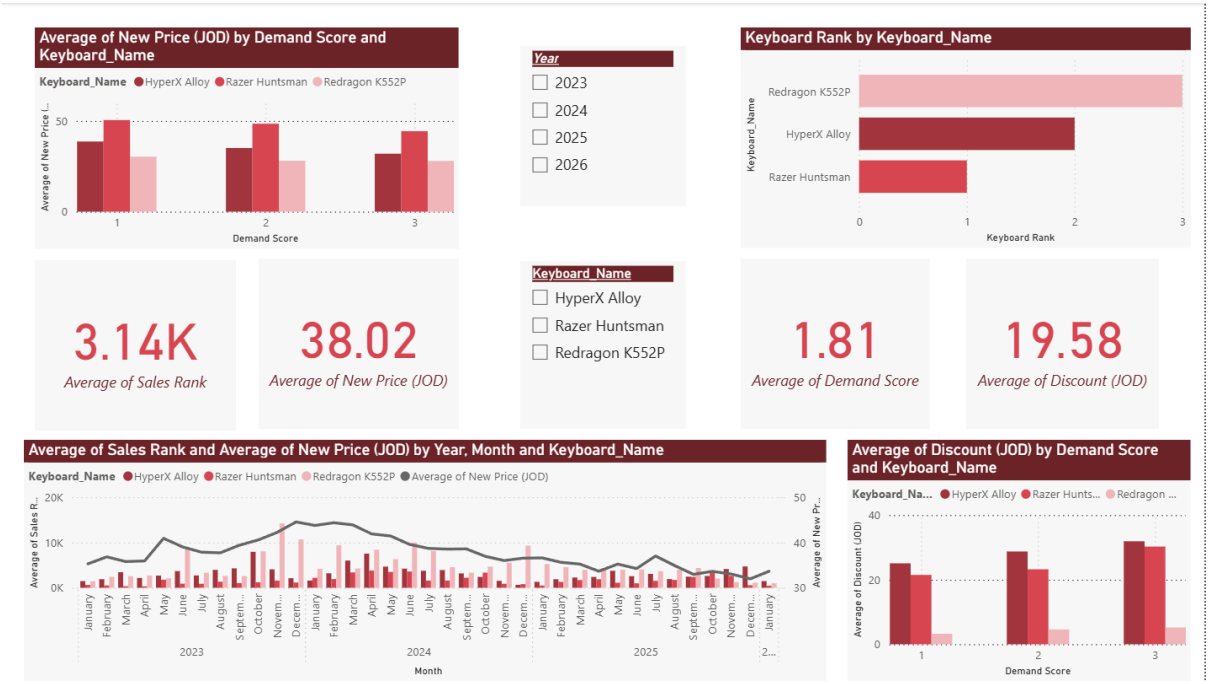
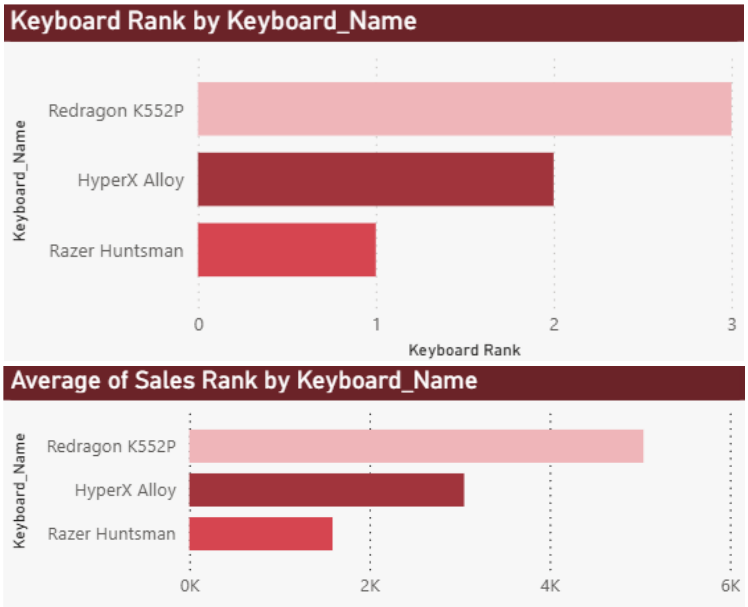


Figure 30: Final Dashboard 2

3.

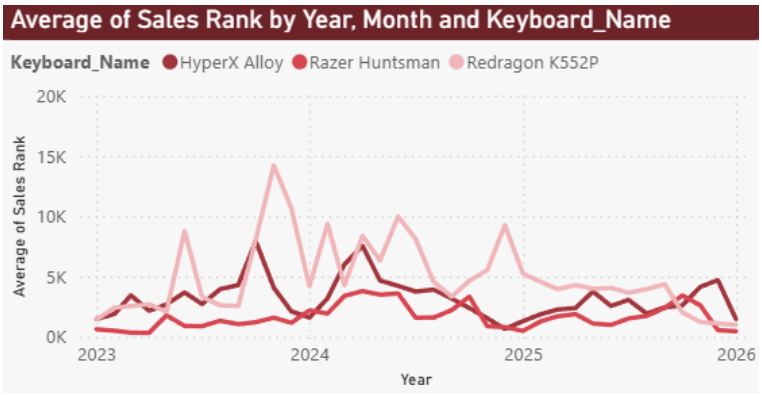
After finalizing the implementation phase, we concluded to some insights like historical analysis, customers demand, future analysis, discount analysis, price analysis, and performance analysis, as we will give City Center recommendations that guide, help, and support their decisions about if they should introduce the three keyboards in the market based on the insights.



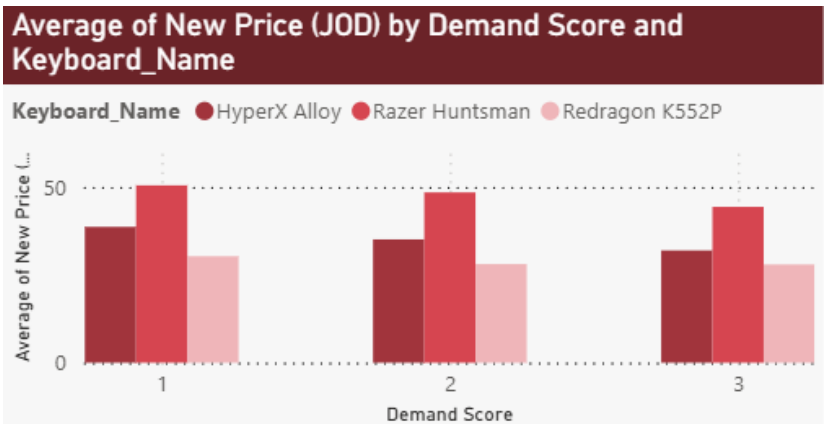
First, from analyzing the ranking for the three keyboards that is shown through the bar chart keyboard rank analysis we can see that Razer Huntsman achieved the highest demand performance across the other keyboards, HyperX Alloy ranked second, and Redragon K552P ranked the lowest.

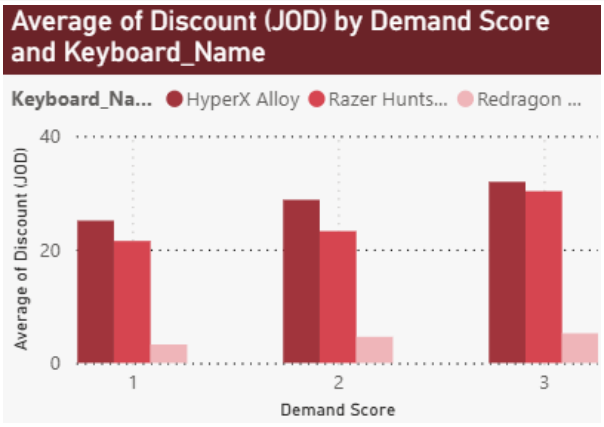
Also, the average sales rank through the bar chart for the three keyboards shows the same results as higher sales rank show lower demand, while lower sales rank shows higher demand, so Razer Huntsman has the lowest sales rank, HyperX Alloy ranked second, and Redragon K552P ranked the lowest.

So, Razer Huntsman shows a strong capability to enter the market, HyperX Alloy shows a moderate capability to enter the market, and Redragon K552P shows a very weak capability to enter the market.



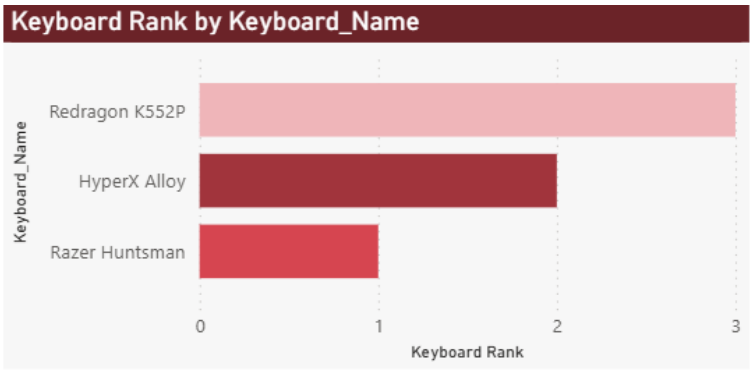
From analyzing the sales rank for the three keyboards by comparison we can see that in terms of stability, Razer Huntsman ranked first with the lowest sales rank and stronger stable demand, HyperX Alloy ranked second with a moderate stability and demand, and Redragon K552P ranked third with a very low stability and very weak demand, so from these insights we can say that in terms of risk, Razer Huntsman show a low risk, HyperX Alloy show a moderate risk, but Redragon K552P shows a very high risk.



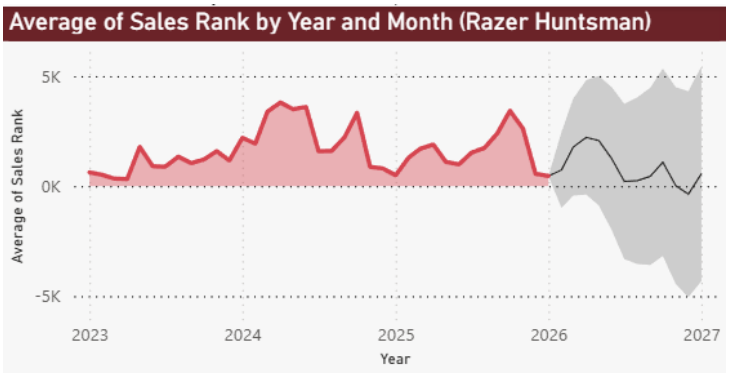


As we can see from the first chart above, we can see the relationship between price and demand score for each keyboard, and how does the brand influence the demand. So we can say that for Razer Huntsman, it stays the highest even if it has the highest prices, but for HyperX Alloy, also shows the second demand even with higher prices than Redragon K552P, and for Redragon, we can say that it always shows lower demand which means that low price does not always show high demand but the key thing is in the brand name itself.

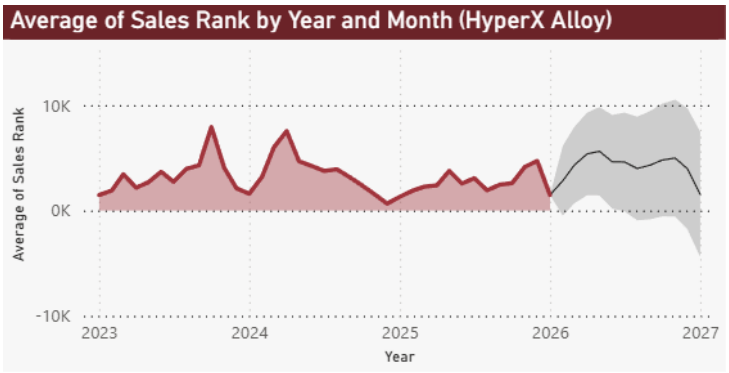
Also, we can say from the second chart that price change or discount affect the demand but not strong effect, so these discounts can help City Center through low peak months so when there is a low demand discount can help a little to increase the demand, but they shouldn't rely on the discount to make demand higher.



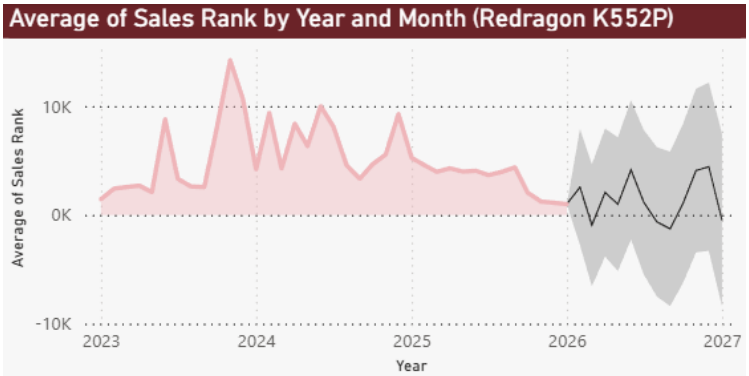
Here we can say that this is the performance ranking as Razer Huntsman > HyperX Alloy > Redragon K552P.



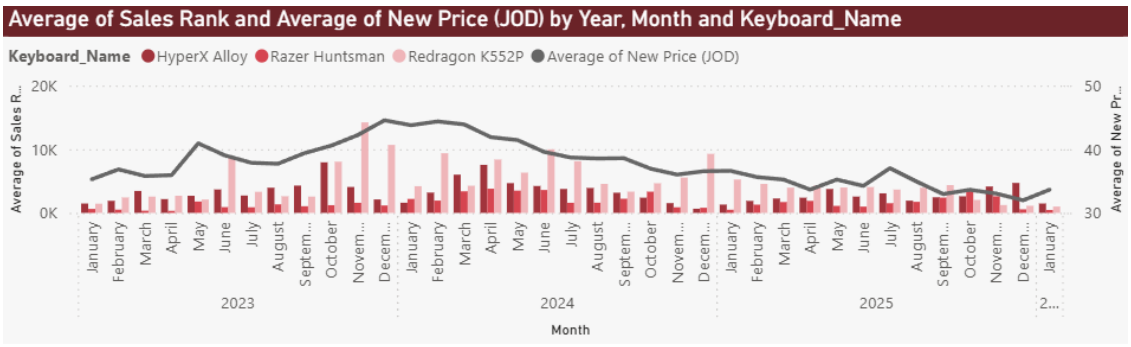
The forecasting results for Razer Huntsman shows that the future for sales rank is has a moderate stability, and it also shows a very good sales rank that indicates a very good demand in the future.



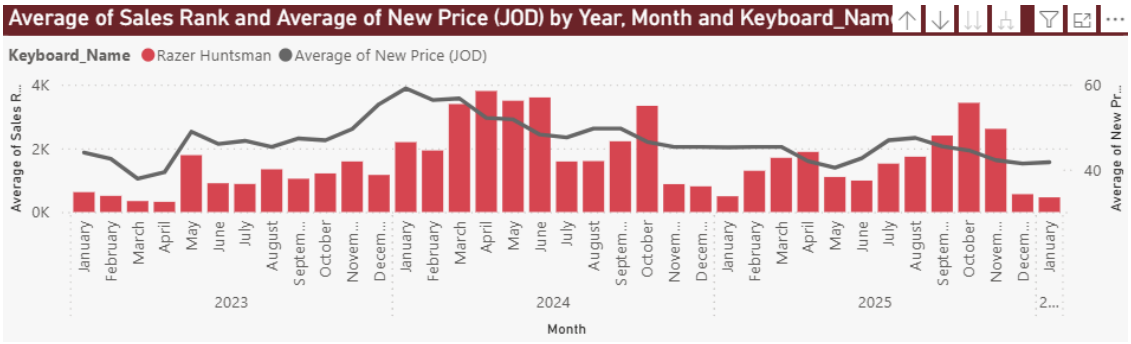
The forecasting results for HyperX Alloy shows that it also has a very low and stable customer demand, and has the highest predicted sales rank compared to Razer and Redragon which means that it may have the lowest demand of all keyboards in the future.



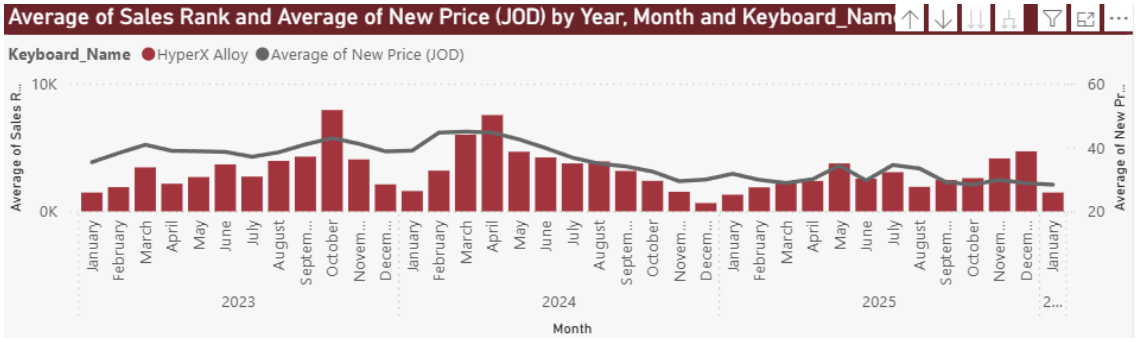
The forecasting results for Redragon K552P also shows that it has a very unstable customer demand with a good predicted sales rank compared to HyperX Alloy which means that it may have a moderate demand of the three keyboards in the future but only on specific months.



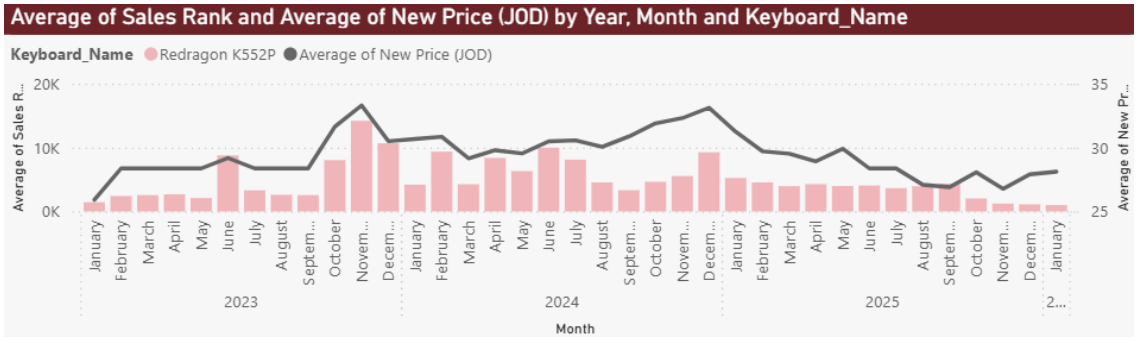
The goal from this chart was to cover at which time and what price we have the highest demand.



For Razer Huntsman, most of the time it has a very strong demand, even at high prices, which shows that Razer brand is so strong.



For HyperX Alloy, most of the time it has a moderate demand, even when prices are high, but sometimes when there is price drop they achieve high demand.



For Redragon K552P, it has the lowest demand across the other two keyboards, even it has the lowest prices, which shows that price alone does not really affect the demand but it is affected with multiple factors such as the brand strength but in 2025 and until now, the Redragon shows a better demand than the previous year which may indicate that the brand or the keyboard is growing.

After Making a full analysis of the implemented solution, we must now as NextStep Solutions make recommendations for City Center to support the decision-making process for introducing the keyboards.

Based on the keyboard rank, average sales rank, and forecasting results for sales rank, we can recommend City Center to introduce Razer Huntsman first, because it shows the lowest average sales rank, and the most stable and high demand over the years and over the future, so Razer Huntsman has the a very low risk for market entry, and a high probability for success.

Then after that, City Center should introduce HyperX Alloy as a second after introducing a little of it and monitor the actual initial sales. Even though the forecasting results weren't so good, but HyperX Alloy proved over the few previous years that it is strong and have a moderate performance compared to the other two keyboards, but if City Center want to introduce this keyboard, they can use some price reduction.

For Redragon K552P, this keyboard has the lowest demand and the lowest stability over the few previous years, so City Center should delay the introduction of this keyboard until its case become clearer, because in the past year, this keyboard showed a good improvement for the demand, so, introducing this keyboard may be risky if things are not very clear.

If City Center want to introduce the three keyboards at once, they must plan their stock inventory, as based on the results, Razer Huntsman should have a high count in the stock inventory, while HyperX Alloy and Redragon K552P should have a low count in the stock inventory because HyperX forecasting results showed that it may have the lowest demand, and because Redragon K552P has proved a poor performance over the past years, so having a moderate or high count of them in the stock inventory may been so risky.

For the discounts, we recommend City Center to use discounts only at low demand months where the demand is not good which can improve the demand a little bit because brand strength affect the demand much more.

For the prices, City Center should introduce Razer Huntsman at a controlled good price because it is proved that Razer Huntsman has high demand even at high prices, but for HyperX Alloy, City Center should introduce it with a medium range price because for that keyboard when there is some small price reduction, the demand is increased a little, but for Redragon K552P, it has a very low reputation so it must be introduced with a low range price, and all of that along with the visuals prove that brand strength has stronger effect on demand than prices.

For the best time to introduce the keyboards, City Center can introduce Razer Huntsman at any time because its demand is stable over the years, but HyperX Alloy should be introduced during the periods where the demand is moderate or high because from the forecasting results, HyperX Alloy demand will go down, but for Redragon K552P it has a good demand over 2024 and 2025 in peak months, so in my opinion this keyboard should be introduced during these month, but it still have a high risk. City Center should not only rely on the forecasting results but also they should rely on other visuals and factors like the performance.

4.

In this section, we will evaluate the overall effectiveness of the applied data science method in meeting the user and business needs of helping the decision-making process for City Center about introducing new keyboards.

From our point of view we see that the applied data science method is very effective to meet the business needs by helping City Center to achieve the overall goal, as they wanted to study three keyboards for introducing them in the market and make a decision by understanding the customer demand, keyboard performance pricing, timing, inventory stock, and overall risk in introducing them.

By using sales data for the three keyboards, we were able to show a strong insights that are understood by City Center team, as we made things easier to understand such as showing the performance and calculate it, showing the perfect time and what is the relationship between demand, pricing, and time, showing average prices, discount, sales rank, and demand for each keyboard, for example, not everyone understands the sales rank because sometimes it is confusing, so we converted sales rank into demand level (high, medium, low), as City Center team were able to easily differentiate between the keyboards that has high or medium or low demand and differentiate the risk between the three keyboards which supports City Center business needs.

Also, by using forecasting for these sales data to predict future sales rank, and not only depending on past data, City Center can easily evaluate the overall future demand and its stability to make decisions about inventory stock for during the next year, which shows that the solution did not only depend on the past , but also strengthen the decisions because there are future insights.

Another thing is that City Center came to NextStep Solutions with a specific problem, and we addressed and answered it successfully and efficiently, by using understandable dashboards and strong charts, City Center now are able to see the problems with its solution.

The solution was also very effective in meeting the user needs, as the users expect from the system to be easy to use, easy to understand, full of useful insights, interactive, and answer the core questions.

The needs of the users were addressed by making clear dashboards that include useful insights as analyzed in the recommendations section, making things simpler and easier through rankings and demand score, and making visual comparisons between the performance of the keyboards, and the solution showed a clear and understandable insights, as we did not do complex and non-understandable visuals, as we did the complex analysis and show them in a way that focuses on decisions outcomes.

Also, we added interactive elements such as slicers to improve the user experience, which helps user to uncover patterns without technical experience which support the user requirements which improved the user experience and independent decision making, as the system helped other users to explore insights alone.

The thing that strengthens the solution more is that we translated complex data into simple business needs insights as we converted the prices from USD to JOD because the pc store is located in Jordan, making sure that the data are preprocessed correctly, and transforming sales rank into demand levels and score which all helped in making the insights more business friendly and reduced the technical complexity.

Also, the combination between predictive analysis, data analysis, and visualization made sure that the decision City Center will make based on truths not only assumptions, and the solution really answers the business question.

Another thing is that the overall solution made things clear about risks, by identifying the risk for each keyboard, City Center can manage inventory resources effectively and reduce trying to sell high risk products.

As we know, every solution comes with limitations or disadvantages, such as that the solution relies on data from Amazon which may not perfect to show the behaviour of Jordanian customers. Also, the forecasting models can be more accurate, as the wizard way is not the perfect forecasting and we can enhance the forecasting and use other factors not only rely on sales rank. The forecasting using wizard way in Power BI sometimes forecast values under zero which other forecasting ways may limit it which is the correct thing, but negative values does not represent demand in reality, but it indicates that the sales rand expected to be very low.

Another limitation is that we focused on sales rank and price columns while there are other factors that affect the demand like ratings and seasonal gaming events that could influence the customer demand, but we did not think of that earlier.

In the future, to solve these limitations or to add alternative methods, we should study and plan for the solution more effectively and give more time to study and plan for the solution, as we could have used data from the Jordanian Market to produce more accurate insights, collect or add more data or columns that are related to the dataset itself, and make predictions and forecasting using much stronger platforms and ways like using python or R programming languages.

Overall, regarding the limitations, the data science solution provided is effective and answers all City Center business questions, as it can support them to make the decision-making process, and the solution is clear, usable, and add value to the business.

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